A finite adaptive Bell strong converse for the amplitude-damping channel

Abstract

For every finite adaptive qubit amplitude-damping protocol, we bound its unconditional Bell overlap in terms of its maximum physical-call count and the sandwiched relative entropy of entanglement of its entire initial Alice–Bob state. The bound uses normalized separable comparisons, includes both damping endpoints, and holds on the output closure at fixed call, logical-dimension, order, and resource budgets. The proof constructs a complete finite phase comparison, controls coherent input-weight records, iterates on the actual adaptive tree, and removes its logarithmic block overhead using finitely many independent copies of the fixed protocol.

The rate threshold in the finite-code bound is

$$C_Z(p) = \log_2[1 + (1-p)p^{p/(1-p)}], \qquad 0 < p < 1,$$

the capacity of the classical Z-channel with damping probability p. For example $C_Z(1/2) = \log_2(5/4) \simeq 0.321928$ bits per call. Below we derive the strong-converse implication for sequences of the finite protocols defined here, including the initial resource charge. The comparison method uses the amortized-entanglement framework [5] and the relative-entropy converse approach [4]. The endpoint value $B_1(p) = \log_2(2-p)$ coincides with the amplitude-damping max-relative entanglement calculated by Rigovacca et al. [3, Proposition 2]; that endpoint is not a new channel bound.

1 States, physical protocols, and the statement

All spaces in the finite construction are finite dimensional over $\mathbb{C}$. A density operator is positive semidefinite with trace one. All logarithms are base two. For $0 \le p \le 1$, put $q = 1-p$ and

$$A_0 = |0\rangle\langle 0| + \sqrt{q}|1\rangle\langle 1|, \qquad A_1 = \sqrt{p}|0\rangle\langle 1|, \qquad \mathcal{A}_p(X) = A_0 X A_0^\dagger + A_1 X A_1^\dagger. \tag{1}$$

The physical call sends a qubit from Alice to Bob and applies this channel; both preexisting memories are acted on by identities. In particular, $\sum_{j=0}^{1} A_j^\dagger A_j = I$.

A finite adaptive protocol is a finite rooted tree. Its nodes are terminal nodes, complete Alice instruments, complete Bob instruments, or physical calls. At an Alice node, matrices $K_j : A \to A'$ satisfy $\sum_j K_j^\dagger K_j = I_A$, and public outcome j selects the continuation. The analogous definition holds for Bob. Dimensions may change from node to node; each continuation has the same prescribed terminal logical spaces. Any additional private Kraus index can be retained locally or refined into finitely many outcomes. Discarding local registers is also a complete local operation. Classical communication in either direction is represented by the outcome-dependent continuations.

Write $P[X]$ for the unconditional terminal operator. Its recursion is

$$\begin{aligned}
P[X] &= X && \text{at a terminal node,}\\
P[X] &= \sum_j P_j[(K_j \otimes I)X(K_j \otimes I)^\dagger] && \text{at an Alice node,}\\
P[X] &= \sum_j P_j[(I \otimes L_j)X(I \otimes L_j)^\dagger] && \text{at a Bob node,}\\
P[X] &= P_1[(\mathrm{id} \otimes \mathcal{A}_p)(X)] && \text{at a call node,}
\end{aligned} \tag{2}$$

where the last expression includes the change of laboratory cut caused by transmission. Its call count $c(P)$ is zero at a terminal node, the maximum of the continuation counts at a local node, and $1 + c(P_1)$ at a call node. Thus $c(P)$ is a maximum over histories, including every failure history. No output is divided by a success probability. Completeness makes P a trace-preserving completely positive (CPTP) map.

For $\alpha > 1$, let $t = (\alpha - 1)/\alpha$ and define

$$\begin{aligned} \widetilde{Q}_\alpha(\rho\|\sigma) &= \mathrm{Tr}[(\sigma^{-t/2}\rho\sigma^{-t/2})^\alpha], \\ \widetilde{D}_\alpha(\rho\|\sigma) &= \frac{1}{\alpha-1}\log\widetilde{Q}_\alpha(\rho\|\sigma), \qquad \operatorname{supp}\rho \subseteq \operatorname{supp}\sigma, \\ \widetilde{E}_\alpha(A:B)_\rho &= \inf_{\sigma\in\mathrm{SEP}(A:B)} \widetilde{D}_\alpha(\rho\|\sigma). \end{aligned} \tag{3}$$

Negative powers act on the support and vanish on the kernel. Unsupported comparisons have divergence $+\infty$. The set $\mathrm{SEP}(A:B)$ consists of finite convex combinations of normalized product densities. In finite dimensions it is closed. The infimum is over *normalized* states. The cut in (3) contains every initial register in Alice's and Bob's laboratories, including resource-correlated side registers.

For $d \geq 2$, write

$$|\Phi_d\rangle = \frac{1}{\sqrt{d}}\sum_{i=1}^{d}|ii\rangle, \qquad \Phi_d = |\Phi_d\rangle\langle\Phi_d|, \qquad G_d(\eta) = \mathrm{Tr}(\Phi_d\eta). \tag{4}$$

This is squared fidelity to the pure Bell target. For $0 < s \leq 1$ define

$$B_s(p) = \begin{cases} 1, & p = 0, \\ 0, & p = 1, \\ \log(2-p), & 0 < p < 1,\ s = 1, \\ \log\left[1 + \left(\dfrac{1-p}{[1-p^{1/(1-s)}]^{1-s}}\right)^{1/s}\right], & 0 < p < 1,\ 0 < s < 1. \end{cases} \tag{5}$$

The endpoint clauses are part of the definition and require no evaluation of the interior quotient at an endpoint.

Theorem 1 (Finite adaptive Bell converse)**.** *Let $0 \leq p \leq 1$, $0 < s \leq 1$, $m \in \mathbb{N}_0$, $d \geq 2$, and $e \in \mathbb{R}$. Let ρ be any density on any finite initial laboratory pair $A:B$, and let P be any finite protocol with terminal spaces $\mathbb{C}^d:\mathbb{C}^d$. If*

$$c(P) \leq m, \qquad \widetilde{E}_{1+s}(A:B)_\rho \leq e, \tag{6}$$

then

$$G_d(P[\rho]) \leq \min\left\{1, 2^{-\frac{s}{1+s}(\log d - e - mB_s(p))}\right\}. \tag{7}$$

For the same fixed p, s, m, d, e, (7) holds for every trace-norm limit of such terminal outputs.

The initial dimensions and the number of local rounds may vary throughout this output family. The condition in (6) is a condition on the entire initial laboratory state. A Bell pair whose two halves are both initially at Alice lies within one laboratory. A shared entangled bank lies across the cut and is included in the bound. The order of the charged quantity is exactly $1 + s$.

For fixed $0 < p < 1$, write $q = 1 - p$. Differentiation at $s = 0$ gives

$$\lim_{s\downarrow 0} \frac{\ln q - (1-s)\ln(1-p^{1/(1-s)})}{s} = \ln q + \frac{p}{q}\ln p.$$

Hence $B_s(p) \to C_Z(p)$. Consider a sequence of finite protocols with call budget $m \to \infty$ and $\log d_m \geq mR$, where $R > C_Z(p)$. Choose a fixed $s > 0$ with $B_s(p) < R$. If the initial resource charges at this order obey $e_m = o(m)$, Theorem 1 makes the Bell overlap tend to zero exponentially. In particular, this applies to separable initial laboratory states, for which $e_m = 0$. This rate conclusion uses exactly the finite-protocol model and output closures stated above.

Comparison with earlier bounds

Let h_2 denote binary entropy in bits. Two analytic benchmarks are the reverse-coherent-information achievable rate

$$L_{\mathrm{RCI}}(p) = \max_{0\leq u\leq 1} [h_2(u) - h_2(pu)]$$

and the balanced-squashing upper bound

$$U_{\mathrm{sq}}(p) = \max_{0\leq u\leq 1} [h_2((1-p/2)u) - h_2(pu/2)].$$

These appear in [6, Supplementary Note 5, Eqs. (S212), (S228)–(S231)]. The achievable entanglement rate is also an achievable secret-key rate. For comparison with the finite-code rate consequence proved above, at $p = 1/2$ the values per physical channel use are

Quantity	Value
Reverse coherent information L_{RCI}	$0.271553\ldots$
Present converse threshold $C_Z = \log_2(5/4)$	$0.321928\ldots$
Balanced-squashing bound $U_{\mathrm{sq}} = \log_2(4/3)$	$0.415037\ldots$
Max-relative bound $\log_2(3/2)$ [3]	$0.584963\ldots$

The first value is the one-variable maximum above. The third is exact: $h_2(3u/4) - h_2(u/4)$ is strictly concave on $(0, 1)$, and its derivative vanishes at $u = 4/9$. The substitution $u = 1/2$ in [6, Eq. (S232)] instead gives $0.410869\ldots$; it does not maximize the preceding expression. Pirandola already identified this issue and evaluated the retained maximization [9, footnote 3 and Theorem IV.1]. Thus $C_Z(1/2)$ improves both analytic upper benchmarks in the table while leaving a gap above the achievable rate. Fawzi, Shayeghi, and Ta also provide a hierarchy of semidefinite strong-converse bounds for assisted quantum capacity [7, Sec. 6]; the table does not rank that hierarchy.

There is a particularly small asymptotic gap at high damping. Abdelhadi et al. give the triple-rail achievable rate $L_3(p) = (1-p)\log_2(3)/3$ [8, Sec. 2.2]. Writing $q = 1 - p \downarrow 0$, the elementary expansion $p^{p/q} = e^{-1}(1 + O(q))$ gives

$$C_Z(p) = \frac{q}{e\ln 2} + O(q^2), \qquad \lim_{p\uparrow 1}\left(\frac{C_Z(p)}{L_3(p)} - 1\right) = \frac{3}{e\ln 3} - 1 = 0.00457489\ldots.$$

The converse threshold is therefore about 0.4575% above this known achievable rate in the high-damping limit. This is a relative asymptotic comparison, not a uniform gap for every p.

2 Order and support estimates

We use the sandwiched-divergence data-processing theorem: for finite CPTP Λ and $\alpha > 1$,

$$\widetilde{Q}_\alpha(\Lambda(\rho)\|\Lambda(\sigma)) \le \widetilde{Q}_\alpha(\rho\|\sigma) \tag{8}$$

on supported positive inputs. The corresponding extended-real divergence inequality holds on all density inputs. This theorem is proved in [1, 2]. It also applies to unnormalized positive inputs: normalize the two inputs separately and use preservation of both traces and the scaling $\widetilde{Q}_\alpha(a\rho\|b\sigma) = a^\alpha b^{1-\alpha}\widetilde{Q}_\alpha(\rho\|\sigma)$.

Lemma 2 (Comparison order)**.** *If $\rho, \sigma, \tau \ge 0$, $\sigma \le c\tau$ with $c > 0$, and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, then*

$$\widetilde{Q}_\alpha(\rho\|\tau) \le c^{\alpha-1}\widetilde{Q}_\alpha(\rho\|\sigma), \qquad \alpha > 1. \tag{9}$$

Proof. Put $t = (\alpha-1)/\alpha \in (0,1)$. The Löwner–Heinz inequality gives $\sigma^t \le c^t\tau^t$. Therefore $C = c^{-t/2}\tau^{-t/2}\sigma^{t/2}$ satisfies $CC^\dagger \le P_\tau$, where P_τ is the support projection, and $\|C\|_\infty \le 1$. The supports obey $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma \subseteq \operatorname{supp}\tau$. Hence

$$\tau^{-t/2}\rho\tau^{-t/2} = c^t C(\sigma^{-t/2}\rho\sigma^{-t/2})C^\dagger.$$

The Schatten-α ideal inequality $\|CXC^\dagger\|_\alpha \le \|C\|_\infty^2\|X\|_\alpha$ proves (9), since $t\alpha = \alpha - 1$. □

Full-rank comparisons on the original input space suffice for every separable infimum used below. Given a supported separable comparison σ and a full-rank product density ζ, set $\sigma_\delta = (1-\delta)\sigma + \delta\zeta$ for $0 < \delta < 1$. Lemma 2 gives

$$\widetilde{D}_\alpha(\rho\|\sigma_\delta) \le \widetilde{D}_\alpha(\rho\|\sigma) - \log(1-\delta). \tag{10}$$

Thus the restricted infimum equals the original one. For full-rank σ there is a finite C_0 with $\rho \le C_0\sigma$. Every CP branch then preserves the necessary support containment. This argument does not require full rank on a subsequently enlarged record space.

Two consequences of (8) will be needed for adaptive recombination. A complete instrument (R_j) gives

$$\sum_j \widetilde{Q}_\alpha(R_j\rho R_j^\dagger\|R_j\sigma R_j^\dagger) \le \widetilde{Q}_\alpha(\rho\|\sigma), \tag{11}$$

by recording the outcome in a direct sum. Discarding a direct-sum flag gives

$$\widetilde{Q}_\alpha\left(\sum_j X_j\middle\|\sum_j Y_j\right) \le \sum_j \widetilde{Q}_\alpha(X_j\|Y_j). \tag{12}$$

These statements include zero-mass branches, with their contributions zero. They do not require normalizing individual outcomes.

3 A complete finite phase comparison

Fix $n \ge 1$. Identify computational basis strings with subsets of $[n] = \{1, \dots, n\}$. Let H_w be the span of $|S\rangle$ with $|S| = w$. For $0 \le r \le w \le n$, define

$$K_L|S\rangle = \begin{cases} \binom{w}{r}^{-1/2}|S \setminus L\rangle, & L \subseteq S,\ |L| = w-r, \\ 0, & \text{otherwise,} \end{cases} \qquad \mathcal{N}_{w,r}(X) = \sum_{|L|=w-r} K_L X K_L^\dagger. \tag{13}$$

Distinct inputs containing a fixed L have distinct survivors. Thus $K_L^\dagger K_L$ is diagonal. Each S contains exactly $\binom{w}{r}$ such loss sets, so $\sum_L K_L^\dagger K_L = I_{H_w}$. The tensor damping map restricted to H_w is exactly

$$\mathcal{A}_p^{\otimes n}\big|_{H_w} = \bigoplus_{r=0}^{w} \pi_{w,r}\mathcal{N}_{w,r}, \qquad \pi_{w,r} = \binom{w}{r} q^r p^{w-r}. \tag{14}$$

The output subspaces H_r are orthogonal. These binomial probabilities sum to one and are independent of the input and all side memories. Zero-probability sectors are retained with zero mass.

Let $\omega = e^{2\pi i/3}$ and let $z \in \{0,1,2\}^n$ be uniform. Define

$$|\chi_z\rangle = \sum_{|S|=w} \omega^{-\sum_{i\in S} z_i}|S\rangle, \qquad |b_z\rangle = \binom{n}{r}^{-1/2} \sum_{|T|=r} \omega^{-\sum_{i\in T} z_i}|T\rangle, \tag{15}$$

and

$$\mathcal{S}_{w,r}(X) = 3^{-n}\sum_z \langle\chi_z|X|\chi_z\rangle |b_z\rangle\langle b_z|. \tag{16}$$

Character orthogonality gives $3^{-n}\sum_z |\chi_z\rangle\langle\chi_z| = I_{H_w}$. Each b_z is normalized. Consequently (16) is a complete finite measure-and-prepare channel. Its output can be placed at Bob using a classical message and local preparation. It preserves separability across the entire laboratory cut with arbitrary old memories.

Lemma 3 (Physical CP domination)**.** *For $0 \le r \le w \le n$,*

$$\mathcal{N}_{w,r} \le_{\mathrm{CP}} c_{n,w,r}\mathcal{S}_{w,r}, \qquad c_{n,w,r} = \frac{\binom{n}{r}}{\binom{w}{r}} \ge 1. \tag{17}$$

The inequality remains valid after tensoring any memory identity.

Proof. Use an unnormalized maximally entangled vector to define Choi matrices. A matrix entry in the Choi vector of $3^{-n/2}|b_z\rangle\langle\chi_z|$, labeled by (S,T), has phase $\omega^{\sum_i z_i(1_S(i)-1_T(i))}$ and magnitude $(3^n\binom{n}{r})^{-1/2}$. Averaging over z separates the signatures $1_S - 1_T$ coordinatewise. The three values $-1, 0, 1$ are distinct modulo three, so this separation has no signature collisions.

All signatures with no negative coordinate have $T \subseteq S$ and are exactly 1_L, where $L = S \setminus T$ and $|L| = w - r$. Their positive rank-one blocks equal the Choi blocks of the K_L in (13), with coefficient $1/\binom{n}{r}$ in place of $1/\binom{w}{r}$. Every other signature contributes a positive block disjoint from these blocks. Multiplication by $\binom{n}{r}/\binom{w}{r}$ therefore leaves the Choi matrix of $\mathcal{N}_{w,r}$ plus a positive remainder. Choi positivity is equivalent to complete positivity, proving (17) and its memory extensions. The cases $w = 0$ and $r = 0$ follow from the same formulas. □

For any supported positive pair X, Y on the input including side memories, Lemma 2, (17), and data processing imply

$$\widetilde{Q}_\alpha(\mathcal{N}_{w,r}(X)\|\mathcal{S}_{w,r}(Y)) \le c_{n,w,r}^{\alpha-1}\widetilde{Q}_\alpha(X\|Y). \tag{18}$$

Here the intermediate comparison is $\mathcal{N}_{w,r}(Y)$, whose support contains that of $\mathcal{N}_{w,r}(X)$; CP domination then supplies the support in $\mathcal{S}_{w,r}(Y)$.

4 Coherent recording and an actual block comparison map

Suppose Alice holds the n qubits to be transmitted. Adjoin a local record W by the isometry $V|S\rangle = |S\rangle|S\rangle_W$, leaving all old memories unchanged. Apply the physical channel to the first factor and call the recorded output Ω_X. Measure W in the binary Fourier basis indexed by $a \in \{0,1\}^n$ and apply the announced phase correction Z^a to Bob's new output qubits. Denote this complete recovery map by $\mathcal{R}$.

$$\mathcal{R}(\Omega_X) = (\mathrm{id} \otimes \mathcal{A}_p^{\otimes n})(X). \tag{19}$$

To verify the identity with arbitrary memory coefficients, expand X in $|S\rangle\langle U|$. A loss set $L \subseteq S \cap U$ gives Fourier factor $2^{-n}(-1)^{a\cdot(S-U)}$. Its surviving strings are $S \setminus L$ and $U \setminus L$, whose phase difference is the same. Bob's correction cancels it. Summing the 2^n outcomes restores each original matrix entry. Thus recovery is an equality of linear maps, including all physical loss terms.

Let $\mathcal{P}$ pinch the local record W by its $n+1$ weight projectors. The weight-w input operators are

$$X_w = (I \otimes \Pi_w) V X V^\dagger (I \otimes \Pi_w), \qquad Y_w = (I \otimes \Pi_w) V Y V^\dagger (I \otimes \Pi_w).$$

Their channel inputs lie in H_w, while the record stays in Alice's side memory. For a separable Y, the Y_w are positive separable operators. Define the recorded comparison and its recovered version by

$$\kappa_Y = \bigoplus_{w,r} \pi_{w,r} \mathcal{S}_{w,r}(Y_w), \qquad \mathcal{T}_{n,p}(Y) = \mathcal{R}(\kappa_Y). \tag{20}$$

The direct sum denotes Alice's orthogonal w record and Bob's orthogonal survivor spaces. Any explicitly attached flags are discarded by complete local maps before recovery. Recording, weight measurement, the finite phase measurement and preparation, and recovery are complete operations. Hence $\mathcal{T}_{n,p}$ is a concrete CPTP separability-preserving map with the same input and output laboratory spaces as the physical block.

Put $\alpha = 1 + s$ and

$$M_{n,s}(p) = \max_{0 \le w \le n} \sum_{r=0}^{w} \pi_{w,r} c_{n,w,r}^s, \qquad F_{n,s}(p) = (n+1)^{1+s} M_{n,s}(p). \tag{21}$$

Both quantities are at least one. Exact direct-sum scaling and (18) give

$$\begin{aligned} \widetilde{Q}_{1+s}(\mathcal{P}\Omega_X \| \kappa_Y) &\le \sum_w \widetilde{Q}_{1+s}(X_w \| Y_w) \sum_r \pi_{w,r} c_{n,w,r}^s \\ &\le M_{n,s}(p) \widetilde{Q}_{1+s}(X \| Y). \end{aligned} \tag{22}$$

The last step is (11) for the recorded weight instrument. The physical and comparison probabilities are matched, so $\pi_{w,r}^{1+s} \pi_{w,r}^{-s} = \pi_{w,r}$ on occupied sectors.

Pinching satisfies $\Omega_X \le (n+1)\mathcal{P}\Omega_X$. The support containments above and this order imply $\operatorname{supp} \Omega_X \subseteq \operatorname{supp} \kappa_Y$. Congruence by $\kappa_Y^{-s/(2(1+s))}$ preserves the order. For positive ordered matrices, each ordered eigenvalue is no larger, so the trace of their $(1+s)$ powers is ordered. Therefore

$$\widetilde{Q}_{1+s}(\Omega_X \| \kappa_Y) \le (n+1)^{1+s} \widetilde{Q}_{1+s}(\mathcal{P}\Omega_X \| \kappa_Y). \tag{23}$$

This uses scalar eigenvalue powers, not operator monotonicity of a power greater than one. Applying recovery data processing to (22)–(23) proves the actual block estimate

$$\widetilde{Q}_{1+s}((\mathrm{id} \otimes \mathcal{A}_p^{\otimes n})(X) \| \mathcal{T}_{n,p}(Y)) \le F_{n,s}(p) \widetilde{Q}_{1+s}(X \| Y). \tag{24}$$

It holds with arbitrary finite side memories and supported positive inputs, including the subnormalized branches of an adaptive protocol.

5 The finite type bound

Define the bit charge

$$c_{n,s}(p) = \frac{1}{s}\log F_{n,s}(p) = \frac{1+s}{s}\log(n+1) + \frac{1}{s}\log M_{n,s}(p). \tag{25}$$

We next prove, throughout $0 \le p \le 1$ and $0 < s \le 1$,

$$c_{n,s}(p) \le nB_s(p) + \frac{2+s}{s}\log(n+1). \tag{26}$$

For interior p, let $h(x) = -x\log x - (1-x)\log(1-x)$, with the continuous endpoint convention, and let $D_{\mathrm{bin}}(v\|q) = v\log(v/q) + (1-v)\log((1-v)/p)$. For $w > 0$, set $u = w/n$ and $v = r/w$. The standard binomial coefficient bound $\binom{N}{k} \le 2^{Nh(k/N)}$ yields

$$\begin{aligned}\pi_{w,r}c^s_{n,w,r} &= \binom{n}{r}^s \binom{w}{r}^{1-s} q^r p^{w-r} \\ &\le 2^{ns[,h(uv)-uh(v)-(u/s)D_{\mathrm{bin}}(v\|q)\,]}.\end{aligned} \tag{27}$$

To see the coefficient bound directly, the kth term of the binomial distribution at parameter k/N is at most one; rearranging gives the inequality, including $k = 0, N$. Both coefficient exponents in (27) are nonnegative. The $w = 0$ sum equals one. Summing at most $n+1$ terms gives $M_{n,s}(p) \le (n+1)2^{ns\beta_s(p)}$, where

$$\beta_s(p) = \max_{0\le u,v\le 1}\{h(uv) - uh(v) - (u/s)D_{\mathrm{bin}}(v\|q)\}. \tag{28}$$

For $v > 0$ put $J(v) = [h(v) + D_{\mathrm{bin}}(v\|q)/s]/v$ and $x = uv$. The expression is $h(x) - xJ(v)$ with $0 \le x \le v$. Its derivative at $u = 0$ is positive infinite, whereas at $u = 1$ it is $\log(1-v) - D_{\mathrm{bin}}(v\|q)/s < 0$, interpreted by a limit at $v = 1$. The unconstrained entropy-conjugate maximizer $x = (1 + 2^{J(v)})^{-1}$ is therefore admissible, and the maximum is $\log(1 + 2^{-J(v)})$. Differentiation gives

$$J'(v) = \frac{\log p + (s-1)\log(1-v)}{sv^2}. \tag{29}$$

For $0 < s < 1$ its unique minimum is $v_s = 1 - p^{1/(1-s)}$, at which $J(v_s) = ((1-s)\log v_s - \log q)/s$. This evaluates (28) as the interior clause in (5). At $s = 1$, J decreases to its minimum at $v = 1$ and gives $\log(1+q)$. At $v = 0$ the original expression is nonpositive, so that boundary cannot increase the maximum. Thus $\beta_s = B_s$, proving (26) for interior p.

At $p = 0$, only $r = w$ has nonzero mass, and $M_{n,s}(0) = \max_w \binom{n}{w}^s \le 2^{ns}$ by the binomial theorem. At $p = 1$, only $r = 0$ has nonzero mass and $c_{n,w,0} = 1$, so $M_{n,s}(1) = 1$. These facts prove (26) with the stated endpoint constants, for every allowed s.

6 Iteration on the complete adaptive tree

Consider a protocol R whose calls are n-qubit physical blocks. Evaluate the *same* tree on a comparison input Y, using the same local instruments and the same outcome-dependent continuations, but replacing each physical block by $\mathcal{T}_{n,p}$. Call the resulting output $R^{\mathrm{cmp}}[Y]$. It is normalized and separable whenever Y is normalized and separable. Support containment propagates by CP branch maps, by (24), and by positive sums.

Lemma 4 (Unconditional tree iteration)**.** *For supported positive X, Y,*

$$\widetilde{Q}_{1+s}(R[X]\|R^{\mathrm{cmp}}[Y]) \le F_{n,s}(p)^{c(R)}\widetilde{Q}_{1+s}(X\|Y). \tag{30}$$

Consequently, if $c(R) \le m$ and $\widetilde{E}_{1+s}(X) \le e$, then

$$\widetilde{E}_{1+s}(R[X]) \le e + mc_{n,s}(p) \tag{31}$$

for density X on the entire input cut.

Proof. Induct on the finite tree. The terminal case is equality. At a local node, (12) first bounds the recombined output quantity by the sum of the continuation quantities. Apply induction to each continuation, replace its exponent by the maximum count using $F_{n,s} \ge 1$, and then apply (11). The resulting factor is exactly $F_{n,s}^{\max_j c(R_j)}$. At a block-call node, induction followed by (24) multiplies the continuation factor by one further $F_{n,s}$. This proves (30) with every branch retained.

Taking logarithms gives $\widetilde{D}_{1+s}(R[X]\|R^{\mathrm{cmp}}[Y]) \le \widetilde{D}_{1+s}(X\|Y) + c(R)c_{n,s}$. For any $\varepsilon > 0$, choose a supported separable comparison with input divergence less than $e + \varepsilon$, using (10) if needed. Its comparison output is a normalized separable state, so the same bound holds for the output infimum. Let ε decrease to zero and use $c_{n,s} \ge 0$ to replace $c(R)$ by m. This proves (31). □

7 Independent finite lanes and the Bell test

Fix the original P, ρ, p, s, m, d, e. For any integer $n \ge 1$, run n independent copies of this same tree and initial state. Pad each history after its last physical call to m calls: prepare a fresh qubit in $|0\rangle$ locally, send it, and discard the received $|0\rangle$ locally. Since $\mathcal{A}_p(|0\rangle\langle 0|) = |0\rangle\langle 0|$, this padding leaves the original output unchanged. It is complete on every history.

Schedule all copies by call number. Before block call j, execute in each lane its own local tree segment up to the next call; then send all n current qubits through $\mathcal{A}_p^{\otimes n}$. This gives a finite m-block protocol retaining every original feedback chronology. Its decoded output is exactly $P[\rho]^{\otimes n}$, including failures. Independent copies of any classical choices are used in distinct lanes.

Tensor products of supported separable comparisons give

$$\widetilde{E}_{1+s}(\rho^{\otimes n}) \le n\widetilde{E}_{1+s}(\rho) \le ne. \tag{32}$$

Indeed functional-calculus powers factor on tensor products, so $\widetilde{Q}_{1+s}(\rho^{\otimes n}\|\sigma^{\otimes n}) = \widetilde{Q}_{1+s}(\rho\|\sigma)^n$; approximate the input infimum. Lemma 4 therefore gives

$$\widetilde{E}_{1+s}(P[\rho]^{\otimes n}) \le ne + mc_{n,s}(p). \tag{33}$$

Across the total two-laboratory cut, $\Phi_d^{\otimes n}$ is a Bell projector of dimension d^n after local relabeling. For any normalized product vector $|a\rangle|b\rangle$, its acceptance is at most d^{-n} by Cauchy–Schwarz. Convexity proves the same bound for every normalized separable comparison, including comparisons with correlations among all lanes inside each laboratory. The actual acceptance is G^n, where $G = G_d(P[\rho])$.

For $G > 0$, binary-test data processing and omission of the nonnegative failure term give, for every separable comparison with acceptance a,

$$\widetilde{Q}_{1+s}(P[\rho]^{\otimes n}\|\sigma) \ge (G^n)^{1+s}a^{-s} \ge G^{n(1+s)}d^{ns}.$$

If $a = 0$, the supported divergence is infinite whenever $G > 0$, so the inequality remains valid. Infimizing gives

$$\widetilde{E}_{1+s}(P[\rho]^{\otimes n}) \geq n \log d + \frac{n(1+s)}{s} \log G. \tag{34}$$

The Bell test is a CPTP map used for a mathematical data-processing bound; it need not be a local measurement within the physical protocol.

Combining (33), (34), and (26), then dividing by n, yields for *every* finite $n \geq 1$

$$\log d + \frac{1+s}{s} \log G \leq e + mB_s(p) + \frac{m(2+s)}{s} \frac{\log(n+1)}{n}. \tag{35}$$

Keep the original code and all its parameters fixed and let n increase. Since $\log(n+1)/n \to 0$, exponentiation gives (7). The extra cap by one follows from density normalization. If $G = 0$, the bound holds directly. This proves the finite part of Theorem 1, including $m = 0$ and both damping endpoints.

8 Fixed-budget output closure

Let $\mathcal{O}_{p,s,m,d,e}$ be the set of all $P[\rho]$ with arbitrary finite initial dimensions and with the conditions (6). Its closure is taken in the trace norm on the fixed terminal space $\mathbb{C}^d \otimes \mathbb{C}^d$. Every member has the same numerical upper bound in (7). If $\eta_j \to \eta$ in trace norm,

$$|G_d(\eta_j) - G_d(\eta)| \leq \|\Phi_d\|_\infty \|\eta_j - \eta\|_1 = \|\eta_j - \eta\|_1 \longrightarrow 0.$$

Positivity and trace one also pass to the limit. Hence (7) holds throughout $\overline{\mathcal{O}_{p,s,m,d,e}}$, proving the closure part of Theorem 1. The common resource bound e remains a condition on every approximating code. This closure specifies a set of finite-dimensional logical outputs; no separate membership theorem for infinite-memory protocols is used.

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